

Progress Report 3

Student: Michael Hein (300375535)

Course: CSIS 4495 – Applied Research Project

Project: Real-Time Cryptocurrency Streaming Pipeline

Reporting Period: Mar 2 – Mar 8, 2026

1. Work Date/Hours Logs

Date	Hours	Description of Work Done
Mar 5, 2026	2.0	Implemented QuickSight setup script (setup_quicksight.py) — creates Athena data source, raw_trades and candles_1m SPICE datasets, and starter analysis with KPI visuals.
Mar 5, 2026	1.5	Configured QuickSight in AWS account — enabled access, granted S3/Athena permissions, and ran setup script. Verified SPICE refresh with live trade data.
Mar 5, 2026	1.0	Updated README with QuickSight setup instructions and added documentation to DocumentsAndReports folder.
Mar 6, 2026	1.5	Updated Athena/QuickSight scripts for improved error handling; resolved cross-region identity issues (us-east-1 vs ca-west-1).
Mar 6, 2026	1.5	Created automated test suite structure: tests/conftest.py , test_config.py , test_producer.py , and test_exchanges.py .

Mar 6, 2026	1.5	Implemented <code>test_sinks.py</code> and <code>test_quicksight.py</code> using mocked boto3 and Kafka libraries.
Mar 7, 2026	2.0	Implemented <code>test_api_server.py</code> — unit tests for arbitrage detection, volume spike detection, and Flask REST endpoints.
Mar 7, 2026	1.0	Resolved pandas <code>FutureWarning</code> by standardizing timezone-aware datetime comparisons in the API server.
Mar 7, 2026	1.5	Integrated and verified Slack webhook for automated alerts (<code>alert_arbitrage</code> and <code>alert_volume_spike</code>).

Total Hours This Period: 12.5

2. Summary Description of Work Done

This reporting period focused on three main areas: (1) Amazon QuickSight BI dashboard setup, (2) automated test suite implementation, and (3) Slack alert integration.

QuickSight BI Integration I implemented an automated setup script to create an Athena data source and two SPICE-backed datasets. The dashboard now includes visual KPIs for trade volume and VWAP trends. A key technical hurdle was QuickSight's requirement for identity operations in `us-east-1` while the project's data resides in `ca-west-1`. I resolved this by adding `QUICKSIGHT_IDENTITY_REGION` logic to the configuration.

Automated Testing A comprehensive `pytest` suite was added to ensure pipeline stability. This covers configuration loading, Kafka producer logic, and exchange adapters (Coinbase, Binance, Kraken). I also fixed pandas deprecation warnings to ensure the pipeline remains compatible with future Python library updates.

Alerting & Monitoring The pipeline's alerting system was successfully connected to Slack via webhooks. Real-time alerts are now triggered whenever the dashboard detects cross-exchange arbitrage opportunities or significant volume spikes.

3. Repo Check-In of Implementation Completed

New Files

- [Implementation/quicksight/setup_quicksight.py](#) – Automation for BI resources.
- [tests/](#) – Full directory containing automated test suite and fixtures.
- [pytest.ini](#) – Global configuration for the testing environment.

Modified Files

- [Implementation/athena/setup_athena.py](#) – Improved partition projection defaults.
- [Implementation/dashboard/api_server.py](#) – Standardized timezone handling for metrics.
- [requirements.txt](#) – Added [pytest](#) and [pytest-cov](#).
- [README.md](#) – Added instructions for QuickSight and Slack alert configuration.

4. AI Use Section

AI Tool Name	Version / Account	Task / Feature	Value Addition
Cursor (Claude)	Claude-3.5-Sonnet / Premium	QuickSight Scripting	Generated boto3 code for SPICE datasets.
Cursor (Claude)	Claude-3.5-Sonnet / Premium	Test Suite Structure	Designed mocks for Kafka and AWS services.
Cursor (Claude)	Claude-3.5-Sonnet / Premium	Alert Verification	Reviewed webhook payload formats and logic.

5. Appendix: AI Prompt History

1. "How do I create a QuickSight data source connected to Athena, and then create SPICE datasets and an analysis using boto3?"
 2. "I'm getting an error regarding user/resource not found in QuickSight. My data is in ca-west-1. Do I need a different region for identity?"
 3. "I need to add a pytest test suite for my crypto pipeline. How should I structure the tests and mock Kafka/AWS?"
 4. "Write unit tests for the api_server module—test arbitrage detection, volume spike detection, and REST endpoints."
 5. "How do I fix FutureWarning regarding timezone-naive datetime comparisons in pandas?"
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